

Provisional Patent Application

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Title of Invention:

VoxTalk: Voice-Activated, AI-Driven Website Assistant with Auto-Trigger Engagement

Background of the Invention:

As artificial intelligence and voice recognition technology continue to evolve, most websites still rely on outdated user interfaces such as contact forms, generic chatbots, or static content. These systems are impersonal, reactive, and often underutilized by site visitors.

There exists a need for a simple, website-integrated, custom-branded, voice-to-voice AI assistant that helps website owners engage visitors immediately — starting the conversation as soon as they arrive on the domain. The system should work for legacy, static, or dynamic websites, and allow businesses to offer customized, voice-guided interaction based solely on their products, services, and internal knowledge.

This invention is designed to serve website owners and their customers, creating a better user experience while protecting brand identity and keeping AI interaction focused on the intended topic.

Summary of the Invention:

This invention provides a method and system for embedding a voice-first, AI-powered assistant into any website. The assistant auto-triggers upon domain load, optionally

beginning with a text-based greeting such as, “Welcome to [Company Name], click here to begin a voice conversation.” Users can then choose to engage using spoken conversation or text, depending on their environment.

The AI system integrates with backend large language models (LLMs) such as OpenAI’s GPT-4, Gemini, Claude, and Perplexity, and it can be trained using the client’s proprietary business information, enabling custom, branded responses. It avoids off-topic discussion and can redirect irrelevant questions back to the business’s focus area (e.g., “Sorry, I’m only trained to assist with Corvette parts”).

Detailed Description:

Core Features:

- **Auto-launch on website load with text-first greeting**
- **Optional voice-to-voice interaction using microphone and speech synthesis**
- **Voice and text toggle for accessibility in quiet or noisy settings**
- **Custom-trained AI models that use the business’s own documents, FAQs, or product catalogs**
- **Controlled response scope to stay on-brand and avoid generic open-ended AI answers**
- **Integration with multiple backend models: GPT-4, Gemini, Claude, and Perplexity**
- **Multi-API support for voice input/output and transcription, including:**
 - **OpenAI (ChatGPT)**
 - **Google Cloud Speech-to-Text**
 - **ElevenLabs for voice synthesis**
 - **WebSockets**
 - **Node.js or Python (Flask/Express) for backend**

- **HTML, CSS, JavaScript for frontend widget**

Technical Workflow:

- 1. User visits website → system auto-triggers with on-screen greeting**
- 2. Option to enable microphone and begin voice conversation**
- 3. Speech captured and transcribed using Google Cloud or Whisper**
- 4. Transcription sent to AI backend (GPT, Gemini, etc.)**
- 5. Response generated based on business-trained data only**
- 6. Output played as synthesized voice using ElevenLabs**
- 7. User may continue speaking or switch to typing**

Use Cases:

- **Auto industry (e.g., classic Corvette parts)**
- **Law firms and medical offices**
- **Local service businesses**
- **Educational or nonprofit sites**
- **eCommerce and online retail**
- **Municipal and tourism websites**

Claims (Sample):

- 1. A system for automatically initiating AI interaction upon website domain load.**

- 2. The system of claim 1, wherein a text prompt is displayed allowing users to opt into voice communication.**
- 3. The system of claim 1, wherein voice interaction uses real-time speech recognition and AI-generated voice replies.**
- 4. The system of claim 1, wherein the backend AI engine is selected from the group consisting of GPT-4, Gemini, Claude, and Perplexity.**
- 5. The system of claim 1, wherein the AI is trained exclusively on business-provided content and rejects off-topic queries.**
- 6. The system of claim 1, wherein the frontend offers both voice and text-based conversation options.**
- 7. A method for deploying the system using HTML, JavaScript, WebSockets, and a Node.js or Python backend.**
- 8. A method for customizing AI responses by uploading proprietary documents to define AI behavior.**